

Tunable electron–flexural phonon interaction in graphene heterostructures

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Peculiar electron–phonon interaction characteristics underpin the ultrahigh mobility¹, electron hydrodynamics^{2–4}, superconductivity⁵ and superfluidity^{6,7} observed in graphene heterostructures. The Lorenz ratio between the electronic thermal conductivity and the product of the electrical conductivity and temperature provides insight into electron–phonon interactions that is inaccessible to past graphene measurements. Here we show an unusual Lorenz ratio peak in degenerate graphene near 60 kelvin and decreased peak magnitude with increased mobility. When combined with *ab initio* calculations of the many-body electron–phonon self-energy and analytical models, this experimental observation reveals that broken reflection symmetry in graphene heterostructures can relax a restrictive selection rule^{8,9} to allow quasielastic electron coupling with an odd number of flexural phonons, contributing to the increase of the Lorenz ratio towards the Sommerfeld limit at an intermediate temperature sandwiched between the low-temperature hydrodynamic regime and the inelastic electron–phonon scattering regime above 120 kelvin. In contrast to past practices of neglecting the contributions of flexural phonons to transport in two-dimensional materials, this work suggests that tunable electron–flexural phonon coupling can provide a handle to control quantum matter at the atomic scale, such as in magic-angle twisted bilayer graphene¹⁰ where low-energy excitations may mediate Cooper pairing of flat-band electrons^{11,12}.

Weak electron–phonon (e–ph) coupling has been used to explain the ultrahigh mobility (μ)¹ and electron hydrodynamics^{2–4} observed in single-layer graphene (SLG) supported on hexagon boron nitride (hBN). The small Fermi surface of the linear Dirac band prevents intravalley electron scattering by longitudinal acoustic (LA) or transverse acoustic (TA) phonons with energies above a low level set by the Bloch–Grüneisen temperature (T_{BG}), whereas the reflection symmetry of ideal SLG prohibits electron scattering by an odd number of flexural (ZA) phonons^{8,9}. Meanwhile, the observed superconductivity⁵ in magic-angle twisted bilayer graphene (MATBG)¹⁰ is thought to be facilitated by e–ph interactions^{11,12} that contribute to a pronounced temperature (T)-dependent resistivity (ρ)¹³. The electronic thermal conductivity (κ_e) offers insight into electron-energy relaxation via e–ph interactions, which is not available from resistivity measurements that probe the electron momentum relaxation. Specifically, inelastic electron coupling with high-energy phonons relaxes the heat current (J_Q) more rapidly than the charge current (J_c) to suppress the Lorenz ratio ($L \equiv \kappa_e \rho / T$) below the elastic electron–impurity (e–i) scattering limit described by the Sommerfeld value, $L_0 = \pi^2 k_B^2 / 3e^2$, where k_B and e are the Boltzmann constant and the elemental charge, respectively¹⁴. In comparison, quasielastic electron scatterings by low-energy phonons restore the Wiedemann–Franz behaviour, $\kappa_e = L_0 T / \rho$.

The Lorenz ratio also sheds light on electron–electron (e–e) interactions. In degenerate graphene, momentum-conserving normal e–e collisions conserve J_c nearly perfectly but do not conserve J_Q (ref. 15). In the hydrodynamic regime where normal e–e scattering dominates e–i and e–ph processes, L of degenerate graphene is suppressed below L_0 . For non-degenerate graphene, the Lorenz physics is flipped. Applied electric fields drive electron–hole counterflow in non-degenerate graphene, where normal e–e collisions do not conserve J_c and contribute to resistivity due to Coulomb attraction. However, normal e–e collisions conserve J_Q in intrinsic graphene and do not contribute to thermal resistivity for the parallel flow of electrons and holes under a T gradient¹⁶. Hence, hydrodynamic electron transport in intrinsic graphene can lead to high κ_e and increase L above the bipolar diffusion limit, as reported in a recent measurement³ that obtained κ_e by neglecting the diminishing e–ph interaction near the Dirac point and at low T . After two decades of graphene experiments and past efforts^{3,17,18}, κ_e has yet to be measured in graphene over a broad T range to probe e–ph interactions.

Here we report a κ_e measurement of exfoliated SLG encapsulated in hBN over the T range from 30 K to 310 K. Instead of neglecting e–ph coupling, our method reveals its effect on κ_e , together with the signature of e–e interactions. By concurrent measurements of κ_e , ρ and the

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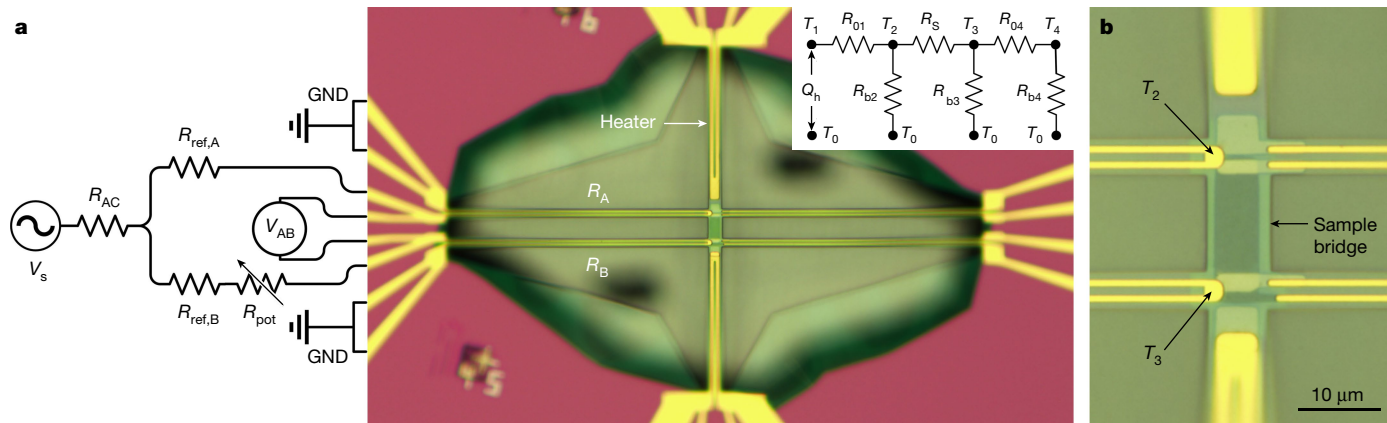


Fig. 1 | The measurement device. **a**, Optical micrograph of the suspended Au/Cr/SiO₂ micro-bridge structure overlaid with a thermal circuit model of the device and a schematic of the Wheatstone bridge used for differential measurement of the variation of the electrical resistances (R_A and R_B) of two U-shape resistance thermometer lines. In the thermal circuit, T_0 is the silicon substrate temperature; T_1 , T_2 , T_3 , and T_4 are the temperatures at the midpoints of the four U-shape

thermometer lines; and R_s , R_{b2} , R_{b3} , R_{b4} , R_{01} , and R_{04} are the thermal resistances between two corresponding temperature points. In the electric circuit, R_{AC} , $R_{ref,A}$ and $R_{ref,B}$ are precision resistors; R_{pot} is a variable resistor; V_s is the voltage output of a lock-in amplifier; and GND is the ground. **b**, Close-up view of the hBN/graphene/hBN/Pt/SiO₂ sample bridge at the centre of the measurement device.

Seebeck coefficient (S), we observe a unusual L peak for degenerate graphene near 60 K, amid a large reduction of L by e–e interactions near 30 K and by inelastic electron scattering by high-frequency phonon polarizations above 120 K. Supported by ab initio calculations of many-body e–ph self-energy and analytical models, our observation suggests that the reflection symmetry of two-dimensional (2D) heterostructures can be manipulated to tune quasielastic electron coupling with an odd number of flexural phonons to control the mobility, L and correlated phenomena.

Probing electronic thermal transport

Figure 1a,b shows an hBN/SLG/hBN heterostructure assembled on top of the centre beam of a suspended silicon dioxide (SiO₂) thin film micro-bridge structure. Four U-shape gold/chromium (Au/Cr) lines on the SiO₂ beams at the two ends of the central sample bridge serve as resistive heaters and thermometers, which are used to obtain the sample thermal resistances shown in the inset of Fig. 1a¹⁹. The conductivity $\sigma = 1/\rho$ (Fig. 2a,b), μ and S are measured with four platinum (Pt) bottom contacts (BCs) in one BC sample (Fig. 1), four Au/Cr edge contacts (ECs) in two samples labelled as EC29 and EC32, and hybrid edge and bottom contacts (HCs) in another sample HC36 (Supplementary Fig. 1). A voltage (V_{gate}) is applied to a Pt gate between the bottom hBN and the SiO₂ bridge to tune the charge concentration $n = n_e - n_h$, where n_e and n_h are the electron and hole densities, respectively. The measured μ (Fig. 2c) increases as the asymmetry between the two graphene/hBN interfaces is reduced by moving the BCs outside the bottom SLG/hBN interface in HC36 and eliminated completely in the two EC samples. The measured μ in the degenerate regime decreases with increasing T from 30 K to 310 K, revealing the effect of e–ph interactions. The electron–hole puddle density (n_{min} ; Fig. 2d) extracted from the σ data at T near 30 K decreases with increasing μ and ranges between $1.1 \times 10^{11} \text{ cm}^{-2}$ in the BC sample and $5.9 \times 10^{10} \text{ cm}^{-2}$ in EC32. The measured S switches sign together with the majority carrier type and increases in magnitude with μ (Fig. 2e,f), consistent with theory²⁰.

Electron–phonon scattering can reduce the lattice thermal conductivity (κ_l) by an amount (κ_l^{e-ph}) that varies with n and σ (ref. 21). As σ is altered by V_{gate} , κ_e and κ_l^{e-ph} are altered by $\delta\kappa_e$ and $\delta\kappa_l^{e-ph}$, respectively. The total change, $\delta\kappa = \delta\kappa_e - \delta\kappa_l^{e-ph}$, results in a variation (δT_2 and δT_3) in the temperatures (T_2 and T_3) at the two ends of the heterostructure, and modifies the electrical resistances, R_A and R_B , of the two corresponding Au/Cr thermometers. This $\delta\kappa$ change is detected by a small signal

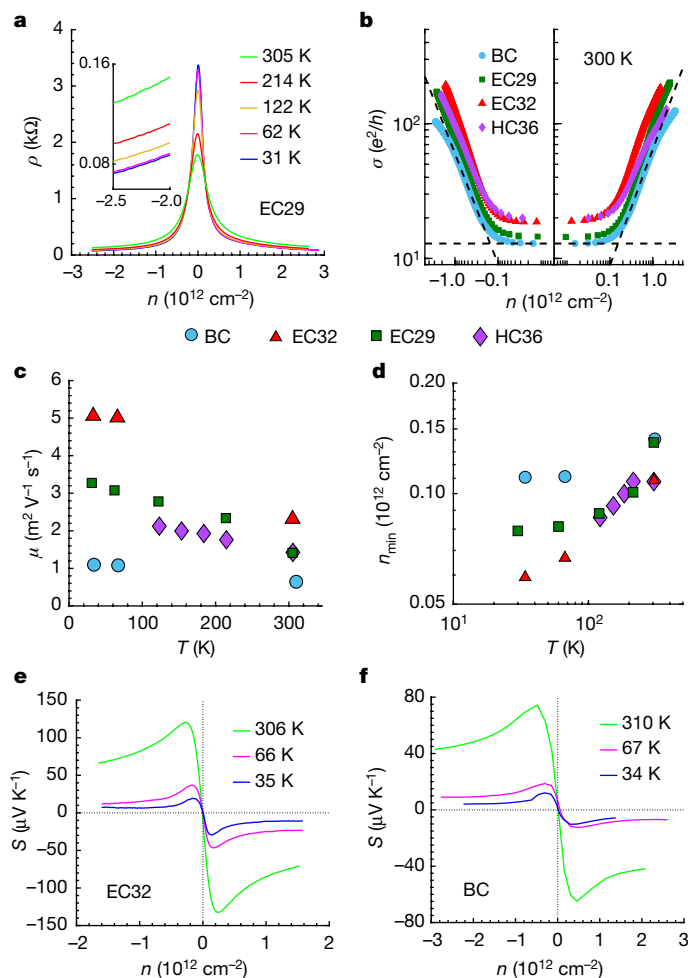


Fig. 2 | Electrical and thermoelectric measurement results. **a**, The resistivity of the EC29 sample. The inset shows the resistivity near the highest hole concentration. **b–d**, Conductivity (**b**), mobility at $1.6 \times 10^{12} \text{ cm}^{-2}$ hole concentration (**c**) and electron–hole puddle density of the four samples (**d**). **e, f**, Seebeck coefficient for the EC32 sample (**e**) and the BC sample (**f**). Panels **a**, **c**, **e** and **f** specify the average graphene sample temperature that is raised by Joule heating in the heater line of the measurement device to be above the sample stage temperature shown in **b** and **d**. In **b**, the black dashed line and dotted line are linear fitting to the BC data in the high- and low- n ranges, respectively, and intercept at a $|n|$ value equal to n_{min} . h is the Planck constant.

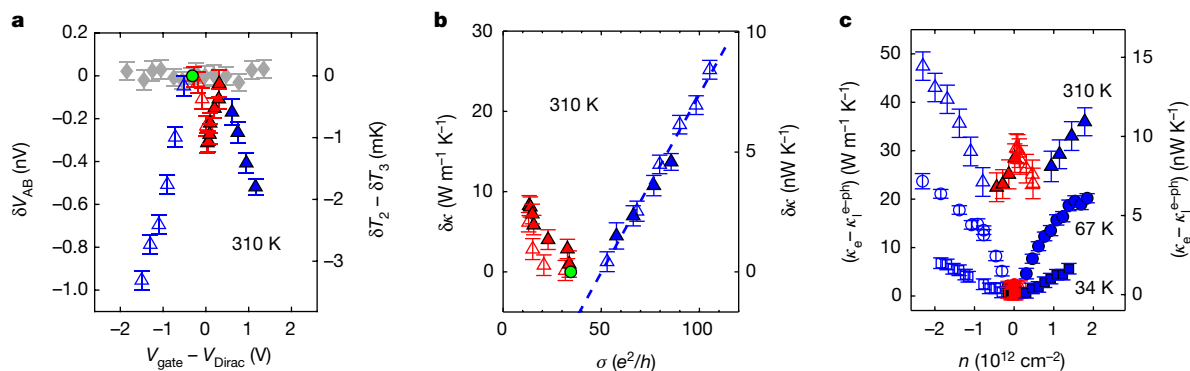


Fig. 3 | Electronic thermal conductivity measurement results of the BC sample. **a**, The measured variation δV_{AB} in the bridge voltage as a function of the gate voltage (V_{gate}) referenced to the Dirac point (V_{Dirac}) and electrical heating rate $Q_h = 79 \mu W$ in the heater line. The average sample temperature is 310 K. The results are grouped into four regions of highly hole-doped (open blue symbols), hole-doped (open red symbols), electron-doped (filled red symbols) and highly electron-doped (filled blue symbols). The green circle is used to highlight the

reference point in the bridge circuit measurement. The grey diamonds show the background signal measured at $Q_h = 0$. The right axis shows the corresponding $(\delta T_2 - \delta T_3)$. **b**, Obtained $\delta\kappa$ as a function of σ at 310 K. **c**, Obtained open-circuit $(\kappa_e - \kappa_l^{e-ph})$ as a function of n at 310 K, 67 K and 34 K. The graphene thickness is taken as 0.335 nm for the κ conversion between the 2D unit of $nW K^{-1}$ and the 3D unit of $W m^{-1} K^{-1}$. The error bars are reported at the 95% confidence level of a two-tailed test, as described in Supplementary Information.

(δV_{AB} ; Fig. 3a) in the Wheatstone bridge voltage of Fig. 1a. The $\delta\kappa(\sigma)$ data in Fig. 3b are analysed to obtain $\kappa_e - \kappa_l^{e-ph}$ in Fig. 3c (Methods), where κ_e is the open-circuit value ($\kappa_{e,o}$)²². The closed-circuit value is calculated as $\kappa_{e,c} = \kappa_{e,o} + S^2\sigma T$, which is expected to approach $L_0\sigma T$ when e–i processes are dominant. The obtained $\kappa_{e,c}$ increases with increased μ (Supplementary Fig. 9), consistent with theory¹⁴.

We calculate $\frac{\kappa_{e,c} - \kappa_l^{e-ph}}{L_0\sigma T} = \frac{\kappa_{e,o} - \kappa_l^{e-ph}}{L_0\sigma T} + \frac{S^2}{L_0}$ and plot it in Fig. 4a,b, where the S^2/L_0 term makes an appreciable contribution only for T above 150 K (Supplementary Fig. 10). For the BC sample at 67 K, this dimensionless number reaches above 0.9 at $|n|$ above $1.5 \times 10^{12} cm^{-2}$, where $\frac{\kappa_l^{e-ph}}{L_0\sigma T}$ is expected to be smaller than 0.1 because inelastic e–ph and e–e processes reduce L below L_0 in degenerate graphene^{14,15}. Based on its dependence on T , n and μ , we expect that $\frac{\kappa_l^{e-ph}}{L_0\sigma T}$ is maximized near the κ_l peak temperature around 100 K and is less than 0.1 for the four samples in the T and n ranges of the measurement, consistent with a theoretical calculation of detectable κ_l^{e-ph} only at $|n|$ well above $1 \times 10^{13} cm^{-2}$ in suspended monolayer and bilayer graphene^{23,24}. Thus, we use the calculated $\frac{\kappa_{e,c} - \kappa_l^{e-ph}}{L_0\sigma T}$ to evaluate the L/L_0 ratio.

Lorenz ratio peak of degenerate graphene

For the BC sample in the degenerate regime, the measured L decreases well below L_0 as T both decreases and increases from about 67 K (Fig. 4b). The L peak magnitude at the intermediate T decreases as μ is increased in the EC29 and HC36 samples and vanishes in the EC32 sample with the highest μ . In contrast, dominant e–e interaction yields a single L minimum at 13 K in the Weyl semimetal tungsten diphosphide²⁵ and at T above 300 K in graphene at $n = 1 \times 10^{12} cm^{-2}$ based on one calculation¹⁵ that ignores e–ph interactions. For simple metals, L exhibits a similar single minimum at an intermediate T owing to inelastic e–ph coupling^{25,26}, whereas it approaches L_0 both at the low T limit because of dominant elastic e–i scattering and at T above the Debye temperature (Θ_D), where the thermal energy ($k_B T$) of Fermi electrons with a wavevector k_F exceeds the phonon energy ($\varepsilon < k_B \Theta_D$) to make e–ph coupling quasielastic.

We note that a previous calculation assuming negligible electron–flexural phonon coupling reports a L peak near 150 K in suspended graphene with n above $1.36 \times 10^{13} cm^{-2}$, corresponding to T_{BG} close to 140 K for the TA modes¹⁴. The calculated L peak is associated with the increasingly quasielastic nature of intravalley electron scattering by TA phonons when the electron thermal energy is increased with T above $k_B T_{BG}$, which sets the upper energy limit of those TA phonons with wavevectors (q) up to $2k_F$ for intravalley scattering. As T increases above

150 K, the calculated L decreases owing to increased electron scattering by high-energy LA and optical phonons in graphene, where Θ_D exceeds 2,000 K to make these scattering processes inelastic.

As n decreases to $1.6 \times 10^{12} cm^{-2}$ in our sample, a reduced T_{BG} to 48 K can in principle lowers the L peak temperature associated with quasielastic electron–TA scattering. However, this T_{BG} decrease is accompanied with a large reduction of electron–TA scattering and a pronounced enhancement of e–e interaction as a result of reduced electron screening (Supplementary Fig. 19)¹⁶. A dominant e–e interaction is expected to reduce L and remove the L peak associated with quasielastic electron–TA phonon scattering. Moreover, electron–TA coupling is not expected to vary substantially among the samples to produce the trend of decreased L peak height with increased mobility. If this trend was caused by increased defect densities in the low-mobility samples, elastic defect scattering would remain strong and keep L close to L_0 at low T without producing the L peak.

Dirac fermion–flexural phonon coupling

This discrepancy has led us to investigate whether the observed L peak could be caused by the neglected flexural phonons. Two previous theories have suggested that flexural phonons dominate the phonon-limited mobility in suspended graphene^{8,9}. By performing many-body ab initio calculations of the phonon-limited electron lifetimes²⁷, we find that electron scattering by a pair of flexural phonons is too weak to yield the measured μ when electron scattering by a single flexural phonon is forbidden by the reflection symmetry in suspended graphene. As reflection symmetry is broken in the hBN/graphene bilayer, the selection rule is relaxed to allow electron scattering by a single flexural phonon to lower the calculated μ below our measured values (Supplementary Fig. 20d), and increases the e–ph scattering rate by over one order of magnitude to be comparable to the e–e quasiparticle scattering rate (Supplementary Fig. 19b). This behaviour survives even when the flexural phonon dispersion is linearized in the long wavelength limit owing to anharmonic renormalization²⁸, stress⁹ or interlayer interactions²⁹ (Supplementary Fig. 15).

Compared with a graphene/hBN bilayer, the density of states per atom for graphene encapsulated in many hBN layers decreases for gapless flexural (ZA) phonons and increases for gapped flexural modes with T_{BG} spanning between 1 K and 150 K. A simplified analytical model shows that this effect can reduce the electron–flexural phonon scattering rate by very different magnitudes depending on hBN thickness and on various structure characteristics that break reflection

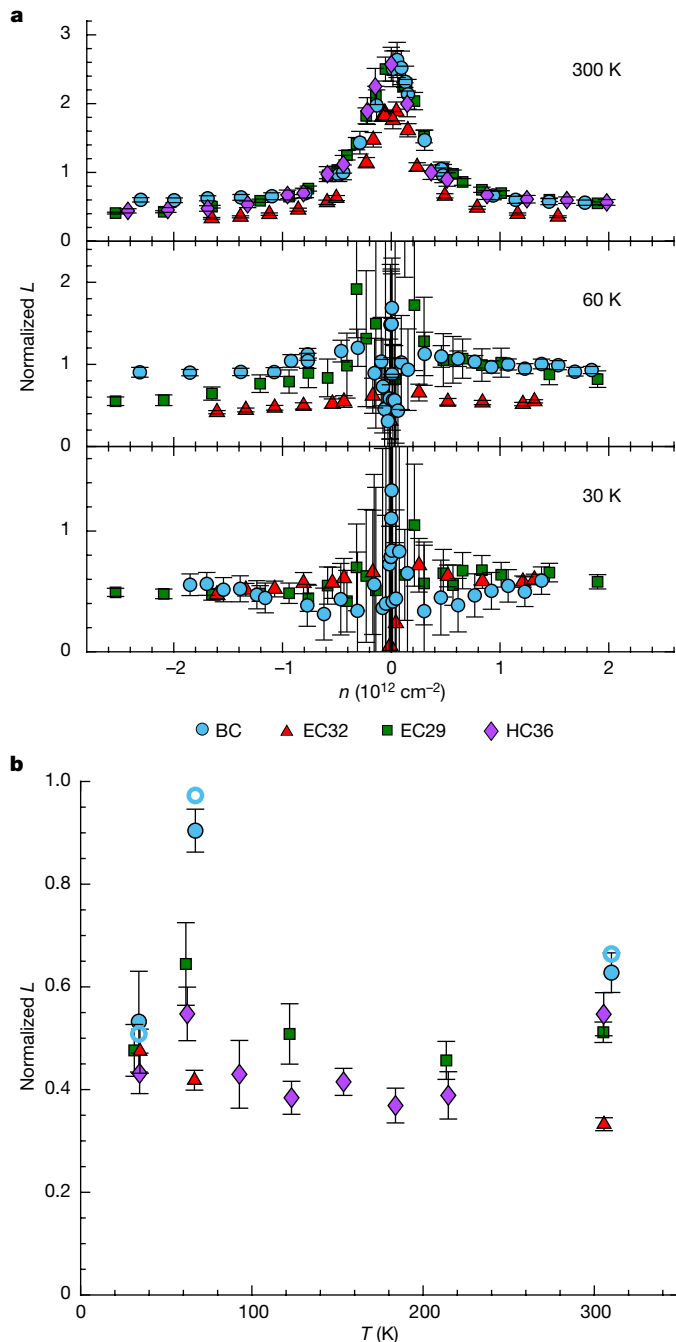


Fig. 4 | Normalized Lorenz ratio. a, b, The normalized Lorenz ratio as a function of n at different sample stage temperatures (a) and sample temperature at a hole concentration of $1.6 \times 10^{12} \text{ cm}^{-2}$ (b). The filled symbols are calculated as $\left(\frac{\kappa_{\text{e,c}} - \kappa_{\text{ph}}}{L_0 \sigma T}\right)$. The open circles in b are calculated as $\left(\frac{1}{T} \frac{\partial(\delta\kappa)}{\partial\sigma} + S^2\right)/L_0$ for the BC sample and are larger than the corresponding error bar. The error bars are reported at the 95% confidence level of a two-tailed test, as described in Supplementary Information.

symmetry. A reduction by about a factor of eight is within the possible range and would increase the calculated mobility to the values measured in the BC sample, and still allows flexural phonon scattering to remain important compared with electron–TA scattering. With energies much lower than the intermediate-wavevector flexural modes that make a large and suppressed contribution to κ_i in suspended and supported graphene³⁰, respectively, these flexural modes with wavevector $q < 2k_F$ exhibit nearly vanishing group velocities and mean free

paths. Consequently, quasielastic electron coupling with these modes increases L and suppresses the mobility and the diffusion thermopower instead of producing detectable phonon-drag thermopower³¹ or κ_i reduction.

Compared with electron–TA phonon interaction, electron coupling with a single flexural phonon is highly sensitive to the sample-to-sample variation of the interlayer interaction and twist angles that break the reflection symmetry, providing a mechanism for the observed decreased L peak height with increased mobility. In these samples, the asymmetry in the two graphene/hBN interfaces are altered by varying the contact design and sample transfer process to reveal a correlation between the asymmetry and both μ and L . The reflection symmetry can also be broken by a twist angle between graphene and hBN in the EC samples to yield non-negligible quasielastic electron–flexural phonon scattering, resulting in the small L peak observed near 62 K for EC29 with an intermediate μ . Although e–i scattering is not weakened at low T , a decreased L of degenerate graphene with T decreased from 70 K to 30 K can be facilitated by a reduced strength of quasielastic electron coupling with flexural and TA phonons (Supplementary Fig. 19b) compared with the still large heat current relaxation rate owing to e–e interaction at 30 K. For EC32 with the highest μ , in comparison, quasielastic electron scattering by either flexural phonons or TA phonons is reduced below both e–e interaction in the T and n ranges and inelastic e–ph coupling at high T to yield a decreased L with increased T from 30 K to 300 K.

Lorenz ratio peak near the Dirac point

As $|n|$ decreases, a very narrow peak in the measured L appears near the Dirac point at near 30 K. However, $|n|$ needs to be lower than a critical value of $n_{\text{cr}} = 5 \times 10^8 \text{ cm}^{-2}$ to achieve a Fermi temperature $T_F < 30 \text{ K}$. As this n_{cr} value is smaller than n_{min} , the non-degenerate range cannot be fully reached at T near 30 K. Thus, L increases only slightly above L_0 near the Dirac point near 30 K despite the increased e–e interactions with decreased $|n|$. As T increases, the L peak near the Dirac point is increased and broadened owing to an increase of n_{cr} to as high as $4.3 \times 10^{10} \text{ cm}^{-2}$ for T near 300 K. At temperatures above 120 K, the L/L_0 peak value lies in between the bipolar diffusion limits for clean SLG and disordered SLG^{3,32}, revealing the dominance of e–ph coupling compared with e–e interaction at temperatures above 120 K.

Conclusions

These results reveal three distinct regimes characterized by hydrodynamic electronic thermal transport at T near 30 K, inelastic electron scattering by high-frequency phonons above 120 K, and tunable quasielastic electron–flexural phonon coupling near 60 K in heterostructure samples with broken reflection symmetry. Although flexural phonons are responsible for high-temperature phonon hydrodynamics in graphene and graphite^{33,34}, electron interactions with these often-ignored low-energy phonons^{13,14,35} can be manipulated at the atomic scale to control transport and quantum states in graphene heterostructures including MATBG. Indeed, recent studies have hinted at the possibility of electron scattering by excitations with lower energies³⁶ and group velocities¹³ than LA or TA phonons being the cause of the observed linear-in- T resistivity in MATBG. This sensitive electronic thermal transport measurement approach can offer insights into tunable e–ph interactions that are essential for the emergence of a variety of correlated phenomena in a broad class of 2D materials and heterostructures.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-05879-y>.

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Methods

Measurement devices

The measurement devices were fabricated on a silicon wafer covered by an approximately 465-nm-thick thermal oxide (SiO_2) for the BC sample and approximately 300-nm-thick SiO_2 for the other three samples. A 10-nm-thick Pt/Cr gate layer was patterned on the oxide film via an electron beam lithography (EBL), electron beam evaporation and lift-off process. The Pt gate appears as the dark green area on the centre sample bridge in Fig. 1b. A 10.8-nm-thick hBN layer was mechanically exfoliated from hBN powder onto another silicon wafer. A hemispherical polydimethylsiloxane handler on a quartz substrate was used to transfer the hBN layer onto the Pt gate layer³⁷. For the BC sample, four 15-nm-thick Pt/Cr bottom contacts were patterned on the transferred hBN layer with the second EBL and lift-off process. The Pt bottom contacts appear in Fig. 1b as the four light moss-green areas at the two ends of the dark green Pt gate. The substrate was annealed at 340 °C in high vacuum for 8 h to minimize polymeric residual and adsorbates. The hemispherical polydimethylsiloxane handler was used to consecutively pick up another 6.4-nm-thick exfoliated hBN flake and an SLG flake exfoliated from a natural graphite source (NGS Naturgraphit, Flake Graphite, 99–99.9% carbon) for the fabrication of the BC sample. The 6.4-nm hBN/SLG heterostructure was subsequently transferred from the handler onto the 10.4-nm bottom hBN layer via a temperature-controlled stamping process in a mask aligner³⁷. The third EBL and a reactive ion etching (RIE) process with CHF_3 chemistry was used to trim the hBN/SLG/hBN heterostructure and expose part of the Pt BCs. The fourth EBL and lift-off procedure was used to pattern 125-nm-thick Au/Cr films into four U-shape Au/Cr resistive heater and thermometer lines and four additional horizontal Au/Cr lines at the right side of the sample bridge in Fig. 1b. These Au/Cr lines are the gold colour lines in Fig. 1b, where the bottom horizontal Au/Cr line is connected to the Pt gate while the other three horizontal Au/Cr lines and the bottom U-shape Au/Cr thermometer line are connected to the four Pt BCs to the graphene. The top U-shape Au/Cr heater line does not overlap with the hBN/SLG/hBN heterostructure. The two horizontal U-shape Au/Cr thermometer lines are electrically isolated from the SLG by the top hBN layer, which appears as the green area that extend under the midpoints, labelled as T_2 and T_3 in Fig. 1b, of the two horizontal U-shape Au/Cr lines. The SiO_2 film was patterned into long beams in the fifth EBL and another RIE process. During this RIE process, the hBN/SLG/hBN heterostructure was etched to have the same width as the centre SiO_2 beam. Tetramethylammonium hydroxide was used to etch the silicon under the SiO_2 beams to suspend the micro-bridge structure.

For the BC sample, raised metal electrodes are present only at the bottom graphene/hBN interface but not at the top graphene/hBN interface, resulting in a larger departure from the reflection symmetry than the other three samples shown in Supplementary Fig. 1. For the HC36 sample with hybrid edge and bottom contacts, the bottom metal contacts were deposited before the transfer and patterning of the bottom hBN at a location outside the bottom graphene/hBN interface, as an attempt to maintain a flat bottom graphene/hBN interface to improve the reflection symmetry. Compared with the BC and HC36 samples, the bottom Pt contact was eliminated for the two EC samples EC29 and EC32 to allow the transfer of the hBN/graphene/hBN heterostructure in one dry-transfer step without either metal or resist residue at the hBN/graphene interface, further improving the symmetry of the two graphene/hBN interfaces. Additional details on the EC measurement devices and measurement methods are provided in Supplementary Information.

Electronic thermal conductivity measurements

When V_{gate} is varied to modify σ from a reference value (σ_{ref}), the total thermal conductivity $\kappa(\sigma)$ of the encapsulated graphene is altered by an amount $\delta\kappa$ compared with the corresponding reference value

$\kappa_{\text{ref}}(\sigma_{\text{ref}})$. Under constant Joule heating (Q_{h}) in the heater line, $\delta\kappa$ results in a change ($\delta T_2 - \delta T_3$) in the temperature drop along the heterostructure. As shown in details in Supplementary Information, the measured electro-thermal bridge voltage (δV_{AB}) can be converted to ($\delta T_2 - \delta T_3$) to demonstrate a temperature sensitivity of 200 μK , which can detect changes as small as 0.01% in the sample thermal resistance (R_{t}). A three-dimensional (3D) numerical heat conduction analysis of the measurement device obtains $\delta\kappa$ from the measured ($\delta T_2 - \delta T_3$) (Supplementary Fig. 6).

We express the lattice thermal conductivity as $\kappa_{\text{l}} = \kappa_{\text{l},0} - \kappa_{\text{l}}^{\text{e-ph}}$, where the second term is used to account for the effect of e-ph coupling on κ_{l} . We further express $\delta\kappa(\sigma) = \kappa_{\text{e}}(\sigma) - \kappa_{\text{e}}(\sigma_{\text{ref}}) - [\kappa_{\text{l}}^{\text{e-ph}}(\sigma) - \kappa_{\text{l}}^{\text{e-ph}}(\sigma_{\text{ref}})]$. In the degenerate regime, both $\kappa_{\text{l}}^{\text{e-ph}}(\sigma)$ and $\kappa_{\text{e}}(\sigma)$ decrease with σ towards zero at $\sigma = 0$. Because free charge carriers cannot be completely removed from graphene to obtain $\sigma = 0$, a polynomial fit of the measured $\delta\kappa(\sigma)$ data of degenerate graphene (Fig. 3b and Supplementary Table 2) is used to extrapolate the intercept $\delta\kappa(\sigma = 0) = \kappa_{\text{l}}^{\text{e-ph}}(\sigma_{\text{ref}}) - \kappa_{\text{e}}(\sigma_{\text{ref}})$. As the charge current in graphene is zero in this measurement, the as-obtained κ_{e} in Fig. 3c is the open-circuit value ($\kappa_{\text{e},0}$)²². According to Fig. 4b, the calculated $\frac{\kappa_{\text{e},0} - \kappa_{\text{l}}^{\text{e-ph}}}{\sigma T}$ value is very close to $(\frac{1}{T} \frac{\partial(\delta\kappa)}{\partial\sigma} + S^2)$ that is calculated directly from the local slope of the linear $\delta\kappa(\sigma)$ data of the BC sample at a hole concentration of $1.6 \times 10^{12} \text{ cm}^{-2}$. Thus, we use the calculated $\frac{\kappa_{\text{e},0} - \kappa_{\text{l}}^{\text{e-ph}}}{\sigma T}$ and $(\frac{1}{T} \frac{\partial(\delta\kappa)}{\partial\sigma} + S^2)$, respectively, to evaluate L in the full n range and only the upper $|n|$ range with a linear $\delta\kappa(\sigma)$ due to a constant L .

Density functional theory calculations

We performed density functional theory calculations using planewaves and pseudopotentials, as implemented in the Quantum ESPRESSO materials simulation suite³⁸, together with the Wannier90³⁹ and EPW codes²⁷. We use the generalized gradient approximation to density functional theory of Perdew, Burke and Ernzerhof⁴⁰ with van der Waals correction, and optimized norm-conserving Vanderbilt pseudopotentials. We set the planewaves kinetic energy cut-offs to 80 rydberg. In the ground-state calculations, we sample the Brillouin zone with Γ -centred uniform meshes of size $72 \times 72 \times 1$. Lattice dynamical properties are computed using a mesh of $12 \times 12 \times 1$ q -points, using density functional perturbation theory. In the self-energy calculations, the Kohn–Sham energies, phonon frequencies and electron–phonon matrix elements are calculated on dense $1,200 \times 1,200 \times 1$ uniform grids using Wannier–Fourier interpolation.

Data availability

Source data are provided with this paper.

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Article

Author contributions L.S. initiated the study. M.M.S. and L.S. designed the experiment for the bottom-contacted sample. Y.H. and L.S. designed the experiment for the edge-contacted and hybrid-contacted samples. E.T. designed the 2D heterostructure transfer method. M.M.S. and Y.H. designed and fabricated the measurement devices, and conducted the measurements, data analysis and numerical simulations for the bottom-contacted and the other three samples, respectively. C.L. and F.G. carried out density functional theoretical calculations. A.H.M. and L.S. developed the analytical models. T.T. and K.W. synthesized the hBN crystals. L.S., M.M.S., Y.H., C.L., F.G., A.H.M. and E.T. wrote the paper with input from T.T. and K.W.

Competing interests The authors declare no competing interests.

Additional information

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